1 (a) A progressive wave is represented by equation

$$y(x,t) = 5\sin 2\pi(2t + 0.05x)$$

where *y* and *x* is the displacement in cm, *t* is time in second. Determine the

(i) direction of wave propagation.

[1 *mark*]

(ii) wavelength

[2 *marks*]

(iii) wave propagation velocity.

[3 *marks*]

(iv) vibrational velocity of particle at x = 15 cm and t = 4 s.

[3 *marks*]

(b) A sinusoidal wave traveling in the +*x* direction (to the right) has maximum displacement of 30 cm, a wavelength of 10 cm and a frequency of 25 Hz. At *t* = 0 s, a particle at equilibrium position. The wave then reflected back along the same path. Write an expression for the standing waves.

[4 *marks*]

(c) In an experiment conducted by a student, a string whose length is 250 cm and a mass of 30 g is used and formed three loops with a frequency of 50 Hz. Determine the wave speed of the string and the tension of the string.

[3 *marks*]

2 (a) Calculate the time it will take to fully charge a car battery if it requires 420 C of charge to pass through it while connected to a charge that delivers a current of 22 A.

[1 *mark*]

(b) A copper wire with a radius of 14 cm is 2 m long and an electric current of 5.6 A flows in the wire. If resistivity of copper wire is $1.7\times10^{-8}\ \Omega\,\mathrm{m}$. Calculate

(i) the resistance of the copper wire.

[2 *marks*]

(ii) potential difference across the cooper wire.

[2 *mark*]

(c)

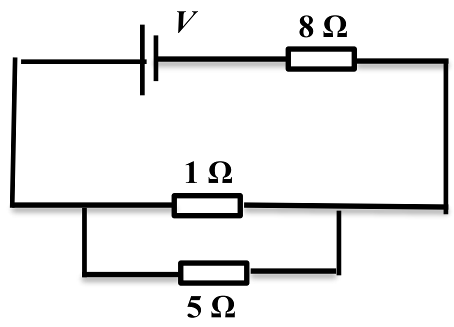

FIGURE 2

FIGURE 2 shows a circuit consists of resistors connected to 12 V battery. Calculate the

(i) effective resistance of the circuit. [3 *marks*]

(ii) total current passes through in the circuit. [2 *marks*]

(iii) potential difference across 5 Ω resistor. [2 *marks*]

3 (a)

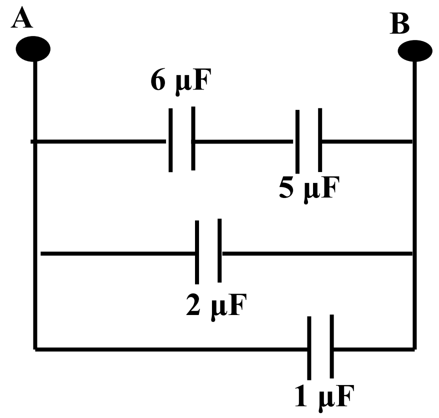

FIGURE 3

FIGURE 3 shows a circuit consists of four (4) capacitors connected between point A and B. If the voltage supply across point A and B is 16 V. Calculate the

(i) effective capacitance of the circuit. [4 *marks*]

(ii) total charge in the circuit. [2 *marks*]

(b) A 60 μF is fully charged by the battery 10 V, then disconnected and discharges through to a 4 k Ω resistor. Calculate the

(i) time constant.

[2 *marks*]

(ii) remaining charge in the capacitor after $t = 0.4$ s.

[2 *marks*]

4

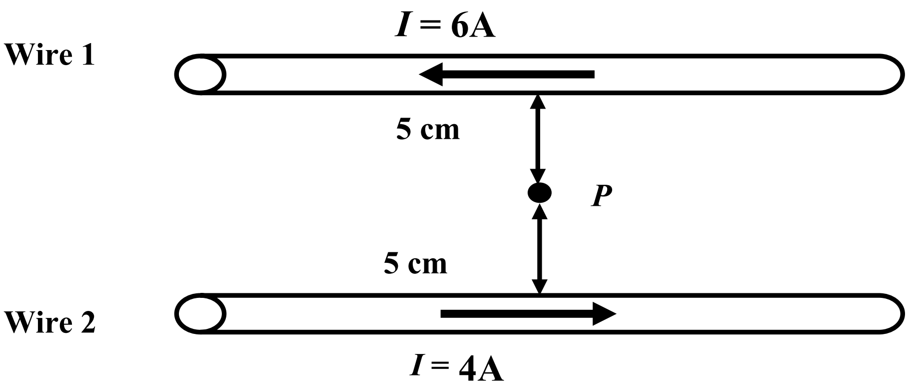

FIGURE 4(a)

(a) **FIGURE 4(a)** shows two long parallel wires, each carrying a 6 A and 4 A current in opposite direction. Determine the magnitude and direction of magnetic field at point P.

(i) due to wire 1

[2 *marks*]

(ii) due to wire 2

[2 *marks*]

(b) A circular loop of surface area 20 cm^2 is placed in a uniform magnetic field of 5 T. Calculate the magnetic flux through the loop when the plane of the coil

(i) is parallel to the magnetic field.

[2 *marks*]

(ii) which has 22 turns and the plane of the coil is making an angle of 60° to the magnetic field.

[3 *marks*]

(c) A metal rod moves perpendicularly through a 2.5 T uniform magnetic field at a speed of 0.02 m s^{-1}. If the length of the rod is 50 cm, calculate the induced emf.

[2 *marks*]

(d)

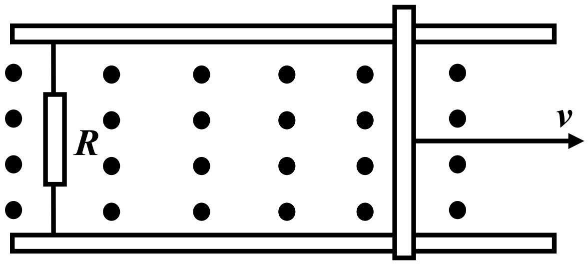

FIGURE 4(b)

FIGURE 4(b) shows a bar moving on rails to the right with a velocity v in a uniform magnetic field directed out of the page. A resistor R connects the rails.

(i) What is the direction of induced current in the rails?

[1 *mark*]

(ii) State **TWO** ways to increase the induced current with R fixed.

[2 *marks*]

5 (a) An object 2 cm high is placed 18 cm from a convex mirror which has a focal length of 6 cm.

(i) Determine the image distance and position of the image.

[3 *marks*]

(ii) Determine the magnification of the mirror.

[2 *marks*]

(iii) Determine the height of the image.

[2 *marks*]

(iv) State **THREE** characteristics of the image.

[2 *marks*]

(b) An object is placed at 60 cm from a converging lens with focal length of 25 cm.

(i) Determine the image distance and position of the image.

[3 *marks*]

(ii) State **THREE** characteristics of the image.

[2 *marks*]

6 (a) Young's double-slit experiment is performed using light of wavelength 590 nm. The distance between the slits and the screen is 1.5 m. The tenth dark fringe is observed 7.25 mm from the central maximum. Calculate the

(i) slit separation.

[3 *marks*]

(ii) distance of the fifth bright from the central maximum.

[3 *marks*]

(b) Young's double slit experiment is performed using 600 nm light. The distance between the slits and the screen is 2.5 m. The eight dark fringe is seen 7.45 mm from the central bright fringe.

(i) Calculate the distance between the two slits.

[1 *mark*]

(ii) What will happen to the interference pattern if the distance between the slits is decreased?

[2 *marks*]

(c) In a Young's double-slits experiment, the slits are 0.25 mm apart and the distance of the screen to the slits is 80 cm. The slits are illuminated by an orange light of wavelength 6.05×10^{-7} m.

(i) Define coherence.

[1 *mark*]

(ii) Calculate the separation between the adjacent bright fringes.

[2 *marks*]

(iii) What is will happen to the interference pattern if a light of shorter wavelength is used instead?

[2 *marks*]